

Mathematics for Machine Learning

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Mathematical Analysis

Limit of a Sequence

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Definition

We call $x \in \mathbb{R}$ the limit of the sequence $\{x_n\}_{n=1}^{\infty}$ if the following condition holds: for each real number $\varepsilon > 0$, there exists a natural number n_0 such that, for every natural number $n \geq n_0$, we have $|x_n - x| < \varepsilon$.

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 - ④ If $x_n \leq y_n$ then $x \leq y$.
- ⑤ If $x_n, y_n \rightarrow x$ and $x_n \leq z_n \leq y_n$, then $z_n \rightarrow x$.

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Theorem

If a sequence of real numbers is decreasing and bounded below then it converges and if it is not bounded below, than its limit is $-\infty$.

Proposition

The sequence $x_n = \left(1 + \frac{1}{n}\right)^n$ is increasing and bounded.

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Theorem

Let x_n and y_n be increasing and decreasing sequences respectively. Also let $x_n < y_n$ for all $n \in \mathbb{N}$ and $(x_n - y_n) \rightarrow 0$. Then the sequences x_n and y_n are convergent and they have the same limit.

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c is called a subsequential limit of the sequence x_n if there exists a subsequence x_{n_k} such that $x_{n_k} \rightarrow c$.

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Every bounded sequence has a convergent subsequence.

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The sequence x_n is called Cauchy sequence if for every $\varepsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that $|x_n - x_m| < \varepsilon$ for all $n, m \geq n_0$.

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Theorem

The sequence x_n is a Cauchy sequence if and only if it is convergent.